

Diploma supplement

Master of Science MSc in Neuro-X

Section:**Neuro-X**

This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO/CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international «transparency» and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

1. Information concerning the holder of the diploma

- | | |
|---------------------------------|------------------------|
| 1.1 Family name(s) | Therras |
| 1.2 First name | Jade |
| 1.3 Date of birth | June 27th, 2001 |
| 1.4 Federal registration number | 19821834 |

2. Information concerning the diploma**2.1 Type of qualification and title conferred with official abbreviation (in original language)**

Qualification	Master of Science MSc
MSc Title	Master of Science MSc in Neuro-X
Swiss Professional title	Ingénieure en Neuro-X (ing. Neuro-X dipl. EPF)

2.2 Main fields of study covered by the qualification

Computational neuroscience, Neural signals processing, Neural interfaces, Translational neuroengineering, Neurosciences, Machine learning

The Neuro-X engineer has a multidisciplinary profile that complements the core competencies of engineers and medical specialists with a strong technological component, making him/her a highly valued professional in the field of neurotechnology. The Neuro-X engineer is apt to become an innovative healthcare professional in industry or hospitals. His/her ability to have a global vision in terms of complex systems, combined with a realistic perspective of what it means to develop a product, make him/her a highly valued professional, capable of interacting with experts from many other fields.

2.3 Name and status of the awarding institution (in original language)

École polytechnique fédérale de Lausanne, Suisse (EPFL)

EPFL - Foundation and Development

Created in 1853 as the École Spéciale de Lausanne, the institution became the Swiss Federal Institute of Technology Lausanne in 1969. EPFL has kept evolving, restructuring its programs and creating new curricula, thus anticipating the scientific and technological developments. Today, EPFL is among the world's top technological universities and a strategic center of innovation in Switzerland.

At a national level, EPFL is expanding by establishing specialized research centers, including associated campus in Neuchatel (Microcity), Sion (Energypolis), Geneva (Campus Biotech), and Fribourg (Smart Living Lab). Each of these research centers is active in a field that is of particular importance to the region where it is located.

EPFL has positioned itself as a research-intensive university of technology, where science and engineering come together to benefit education as well as fundamental and applied research.

EPFL Facts - 2024

- 14'012 students, including 2'401 doctoral students, representing over 120 nationalities
- 6'476 employees, including 377 faculty members and 3'677 scientific collaborators
- 1'369 Master and 434 Doctorate degrees were delivered
- 441 laboratories, 74 research groups
- 107 Start-ups, 28 Corporate companies, 7 SME and 10 Service providers representing a total of 2'885 employees at EPFL Innovation Park
- 1'400'000 visitors to the Library, collections with 245'000 volumes; campus-wide online access to 30 databases, over 24'000 scientific journals and over 170'000 ebooks
- a total budget in excess of CHF 1 billion

Study Programs

- School of Architecture, Civil and Environmental Engineering (ENAC): Architecture; Civil Engineering; Environmental Sciences and Engineering
- School of Engineering (STI): Electrical and Electronic Engineering; Mechanical Engineering; Materials Science and Engineering; Microengineering; Robotics; Energy Science and Technology
- School of Computer and Communication Sciences (IC): Computer Science; Communication Systems; Data Science; Cyber Security
- School of Basic Sciences (SB): Physics; Applied Physics; Mathematics; Applied Mathematics; Molecular and Biological Chemistry; Chemical Engineering and Biotechnology; Computational Science and Engineering; Nuclear Engineering; Statistics; Chemistry and Chemical Engineering
- School of Life Sciences (SV): Life Sciences Engineering
- College of Management of Technology (CDM): Management, Technology and Entrepreneurship; Financial Engineering, Sustainable Management and Technology
- College of Humanities (CDH): Digital Humanities

The EPFL programs are certified by the Swiss agency for accreditation and quality assurance (AAQ), and those in engineering have also been certified by the French Commission des Titres d'Ingénieur (CTI). Thanks to this double certification, EPFL degrees receive the label EUR-ACE and are recognized throughout Europe.

Technology transfer

EPFL acts as a business incubator, ensuring the transfer of academic results to industrial developments. In recent years, EPFL scientists have launched an average of one new company every fortnight. EPFL Innovation Park, which is home to a community of start-ups, scale-ups and innovation units from large companies, is expanding to new sites in order to meet the changing needs of companies. The campuses in Ecublens, Geneva (Campus Biotech EPFL Innovation Park) and Prilly (Unlimitrust) are home to more than 200 companies, including around thirty major international groups and more than 170 start-ups, SMEs and service companies.

International Relations

While preserving the coherence of rigorous scientific training, EPFL offers its students exchange possibilities thanks to agreements with more than 200 carefully selected partner universities worldwide. EPFL is also a member of several academic networks, such as EuroTech, RESCIF and CESAER, which promote scientific collaborations as well as student and faculty mobility.

2.4 Name and status of institution (if different from 2.3) administering studies

Some courses in complementary fields can be chosen at the University of Lausanne.

2.5 Languages of instruction/examination

In English

3. Information concerning the level of qualification**3.1 Level of qualification**

Master: Academic degree obtained after the second cycle of studies.

3.2 Official duration of studies

The Master comprises two stages:

- a. "Master cycle of 3 semesters = 90 ECTS "
- b. Master project of 1 semester = 30 ECTS credits

1 credit = 25-30 hours of workload

3.3 Admission requirements

EPFL Bachelor's degree in Life Sciences Engineering, Electrical and Electronic Engineering, Micro-engineering or Communication Systems or according to qualifications, for other diplomas considered equivalent.

4. Information concerning programme content and results gained**4.1 Mode of study**

Full-time

4.2 Programme requirements

The Master project is carried out after the 90 ECTS credits have been obtained in the Master cycle.

4.3 Programme details and individual grades/marks/credits obtained

The marks and the credits obtained appear in the grade sheet.

4.4 Grading system and if available information concerning allocation of grades

Marking system EPFL

Excellent	6.0
Very good	5.5
Good	5.0
Satisfactory	4.5
Pass	4.0
Fail	3.5
Poor	3.0
Poor to very poor	2.5
Very poor	2.0
Not measurable	1.0

4.5 Overall classification of the qualification

Final Master grade: 5.68 / 6**5. Information concerning the function of the qualification**

5.1 Access to further studies

Allows application for specialised Master and other higher education programmes, in particular Doctoral programmes.

5.2 Professional status

There are no specific regulations bound to this profession

6. Additional information

6.1 Additional details on the individual course of the studies:

Bachelor's degree:

Master project out of EPFL:

Title of the Master project in the original language:

Bachelor of Science BSc in Life sciences engineering
Harvard Medical School , Boston, Massachusetts, U.S.,
United States of America

Investigating inertial motion capture for monitoring gait
differences: A case study on simulating ankle-foot
orthoses with a novel active device in stroke survivors

6.2 Others sources of information

Tel:
Web:
E-mail:

Registrar's office

EPFL AVP-E SAC
Bâtiment polyvalent (BP)
Station 16
CH-1015 Lausanne
Tél: + 41 (0)21 693 43 45
Web: <https://www.epfl.ch/education/studies/>
E-mail: student.services@epfl.ch

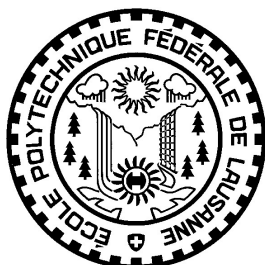
ENIC network: <http://www.enic.ch>
swissuniversities: <http://www.swissuniversities.ch>

7. Endorsement of the Diploma Supplement

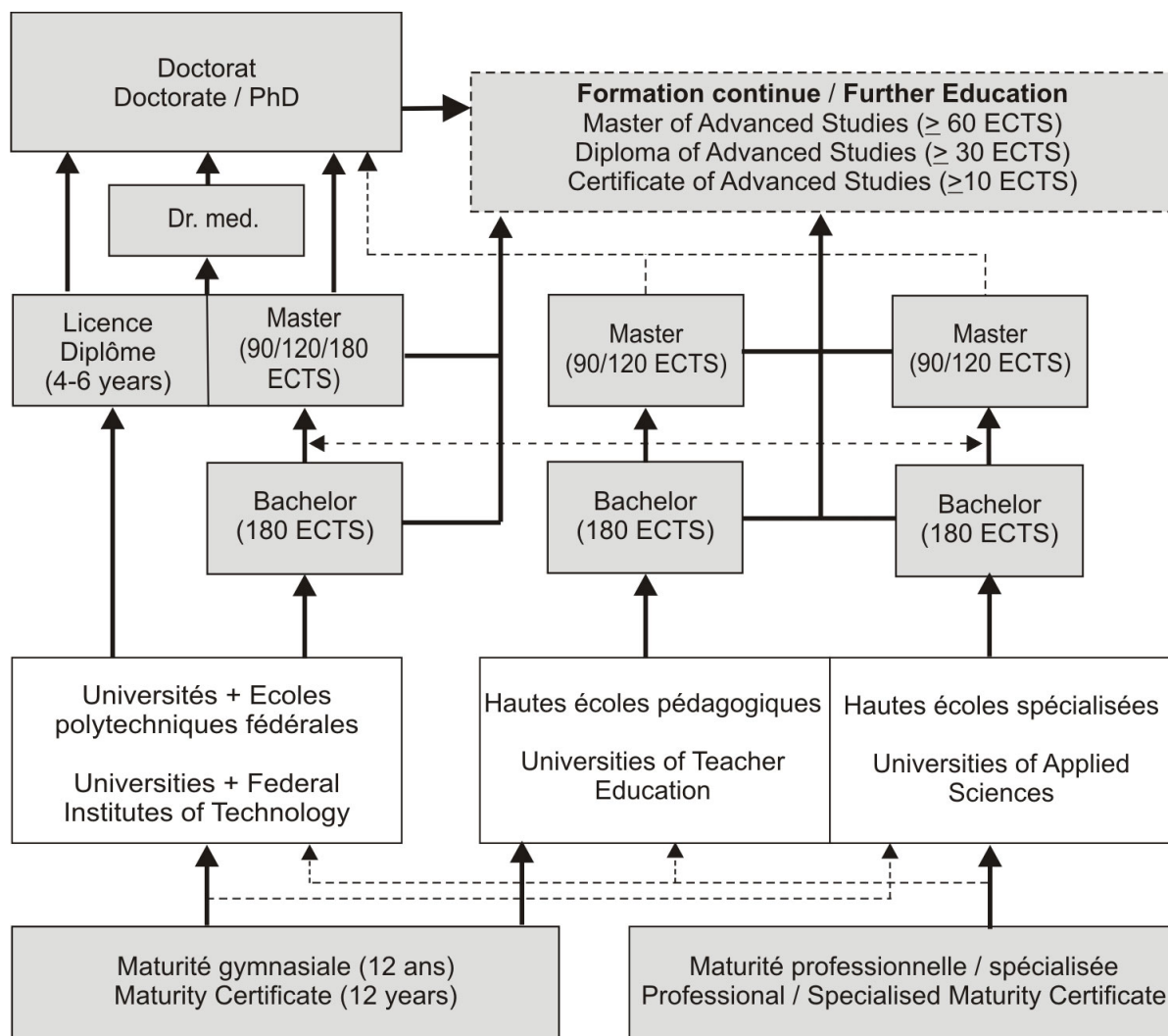
September 25th, 2025

Associate Vice President for Education

Professor Nicolas Grandjean



8. Information on the Swiss education system



9. Description of courses attended**BIO-483 Neuroscience: behavior and cognition**

ECTS Credits : 5.

Lecturers : Blanke Olaf, Herzog Michael, Sandi Perez Maria del Carmen. Semester : 2022-2023 Master semester 2

The goal is to guide students into the essential topics of Behavioral and Cognitive Neuroscience. The challenge for the student in this course is to integrate the diverse knowledge acquired from those levels of analysis into a more or less coherent understanding of brain structure and function.

BIOENG-404 Analysis and modelling of locomotion

ECTS Credits : 4.

Lecturers : Aminian Kamiar, Courtine Grégoire, Ijspeert Auke. Semester : 2022-2023 Master semester 2

The lecture presents an overview of the state of the art in the analysis and modeling of human locomotion and the underlying motor circuits. Multiple aspects are considered including neurophysiology, gait characterization, biomechanics, numerical modeling, neuroprosthetics, and links to biped robots

CS-401 Applied data analysis

ECTS Credits : 8.

Lecturers : Brbic Maria. Semester : 2024-2025 Master semester 3

This course teaches the basic techniques, methodologies, and practical skills required to draw meaningful insights from a variety of data, with the help of the most acclaimed software tools in the data science world (pandas, scikit-learn, Spark, etc.)

CS-433 Machine learning

ECTS Credits : 8.

Lecturers : Flammarion Nicolas Henri Bernard, Jaggi Martin. Semester : 2023-2024 Master semester 1

Machine learning methods are becoming increasingly central in many sciences and applications. In this course, fundamental principles and methods of machine learning will be introduced, analyzed and practically implemented.

CS-444 Virtual reality

ECTS Credits : 4.

Lecturers : Boulic Ronan. Semester : 2022-2023 Master semester 2

The goal of VR is to embed the users in a potentially complex virtual environment while ensuring that they are able to react as if this environment were real. The course provides a human perception-action background and describes the key programming techniques for achieving efficient VR applications

DH-415 Ethics and law of AI

ECTS Credits : 4.

Lecturers : Rochel Johan Robert. Semester : 2023-2024 Master semester 1

This master course enables students to sharpen their proficiency in tackling ethical and legal challenges linked to Artificial Intelligence (AI). Students acquire the competence to define AI and identify ethical and legal questions linked to its conception and increased use in society.

HUM-441 Collective creation: improv-arts & engineering I

ECTS Credits : 3.

Lecturers : Henein Simon Nessim. Semester : 2023-2024 Master semester 1

This course contrasts improvisation in the performing arts with engineering design. Creative processes will be compared and student projects will culminate in an improvised public performance evaluated by the interplay of their artistic and technical creations.

HUM-443 Collective creation: improv-arts & engineering II

ECTS Credits : 3.

Lecturers : Henein Simon Nessim. Semester : 2023-2024 Master semester 2

This course contrasts improvisation in the performing arts with engineering design. Creative processes will be compared and student projects will culminate in an improvised public performance evaluated by the interplay of their artistic and technical creations.

ME-428 Data-driven design & fabrication methods

ECTS Credits : 3.

Lecturers : Hughes Josephine Anna Eleanor. Semester : 2023-2024 Master semester 2

There is an increasing need for data-driven methods for automated design and fabrication of complex mechanical systems. This course covers methods for encoding the design space, optimization and search approaches and digital fabrication methods.

ME-516 Lifecycle performance of product systems

ECTS Credits : 3.

Lecturers : Priot Damien Antoine. Semester : 2023-2024 Master semester 2

Provide the conceptual, technical and methodological understanding of measuring and evaluating the production/consumption decisions on the environmental, social and economic performance of products and services over their life cycle.

MICRO-553 Haptic human robot interfaces

ECTS Credits : 4.

Lecturers : Bourri Mohamed, Shokur Solaiman. Semester : 2023-2024 Master semester 2

This course teaches basic knowledge on haptic devices, force feedback and mechanical man-machine interfaces. Lectures are about 30 %, the rest is hands-on practical work with the "haptic paddle", a complete mechanical device with full laptop control interface. Realization of project in groups of 2.

MSE-341 Sustainability and materials

ECTS Credits : 3.

Lecturers : Abitbol Tiffany. Semester : 2023-2024 Master semester 2

The aim of the course is to provide an overview of sustainability issues as they relate to materials science.

NX-414 Brain-like computation and intelligence

ECTS Credits : 4.

Lecturers : Mathis Alexander. Semester : 2022-2023 Master semester 2

Recent advances in machine learning have contributed to the emergence of powerful models for how humans and other animals reason and behave. In this course we will compare and contrast how such brain models as well as brains create intelligent behaviour.

NX-421 Neural signals and signal processing

ECTS Credits : 6.

Lecturers : Micera Silvestro, Van De Ville Dimitri Nestor Alice. Semester : 2024-2025 Master semester 3

Understanding, processing, and analysis of signals and images obtained from the central and peripheral nervous system

NX-422 Neural interfaces

ECTS Credits : 6.

Lecturers : Lacour Stéphanie, Shoaran Mahsa. Semester : 2023-2024 Master semester 1

Neural interfaces (NI) are bioelectronic systems that interface the nervous system to digital technologies. This course presents their main building blocks (transducers, instrumentation & communication), reviews current and upcoming materials and technological solutions for implantable & wearable NI

NX-423 Translational neuroengineering

ECTS Credits : 6.

Lecturers : Blanke Olaf, Courtine Grégoire, Hummel Friedhelm Christoph, Micera Silvestro. Semester : 2022-2023 Master semester 2
This course integrates knowledge in basic, systems, clinical and computational neuroscience, and engineering with the goal of translating this integrated knowledge into the development of novel methods, technology for the clinical application for patients suffering from neuropsychiatric disorders.

NX-491 Neuro-X project I

ECTS Credits : 8.

Lecturers : Micera Silvestro. Semester : 2023-2024 Master semester 2

The student applies the acquired skills to an academic project.

NX-492 Neuro-X project II

ECTS Credits : 7.

Lecturers : McCabe Brian Donal. Semester : 2023-2024 Master semester 1

The student applies the acquired skills to a scientific, industry or a clinical project.

NX-599 Master project in Neuro-X

ECTS Credits : 30.

Lecturers : Micera Silvestro. Semester : 2024-2025 Master Project spring

Students apply the scientific and technical knowledge they have acquired during their studies to a research case study in an independent way.

Diploma supplement

Bachelor of Science BSc in Life sciences engineering

Section:

Life Sciences Engineering

This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO/CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international «transparency» and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

1. Information concerning the holder of the diploma

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| 1.1 Family name(s) | Therras |
| 1.2 First name | Jade |
| 1.3 Date of birth | June 27th, 2001 |
| 1.4 Federal registration number | 19821834 |

2. Information concerning the diploma

2.1 Type of qualification and title conferred with official abbreviation (in original language)

Qualification	Bachelor of Science BSc
BSc Title	Bachelor of Science BSc in Life sciences engineering

2.2 Main fields of study covered by the qualification

Basic Sciences, Basic knowledge of the various branches of Life Sciences and engineering.

2.3 Name and status of the awarding institution (in original language)

École polytechnique fédérale de Lausanne, Suisse (EPFL)

EPFL - Foundation and Development

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At a national level, EPFL is expanding by establishing specialized research centers, including outposts in Neuchâtel (Microcity), Sion (Energypolis), Geneva (Campus Biotech), and Fribourg (Smart Living Lab). Each of these research centers is active in a field

that is of particular importance to the region where it is located.

EPFL has positioned itself as a research-intensive university of technology, where science and engineering come together to benefit education as well as fundamental and applied research.

EPFL Facts - 2021

- 12'720 students, including 2'407 doctoral students, representing over 120 nationalities
- 6'389 employees, including 346 faculty members and 3'742 scientific collaborators
- 1'175 Master and 437 Doctorate degrees were delivered in 2021
- 413 laboratories and research chairs
- 119 Start-ups, 26 Corporate companies, 8 SME and 17 Service providers representing a total of 2'663 employees at EPFL Innovation Park
- 640'000 visitors to the Library, collections with 275'000 volumes; campus-wide online access to 42 databases, over 26'000 scientific journals and over 130'000 ebooks
- a total budget in excess of CHF 1 billion

Study Programs

- School of Architecture, Civil and Environmental Engineering (ENAC): Architecture; Civil Engineering; Environmental Sciences and Engineering
- School of Engineering (STI): Electrical and Electronic Engineering; Mechanical Engineering; Materials Science and Engineering; Microengineering; Robotics; Energy Science and Technology
- School of Computer and Communication Sciences (IC): Computer Science; Communication Systems; Data Science; Cyber Security
- School of Basic Sciences (SB): Physics; Applied Physics; Mathematics; Applied Mathematics; Molecular and Biological Chemistry; Chemical Engineering and Biotechnology, Computational Science and Engineering; Nuclear Engineering; Statistics; Chemistry and Chemical Engineering
- School of Life Sciences (SV): Life Sciences Engineering
- College of Management of Technology (CDM): Management, Technology and Entrepreneurship; Financial Engineering, Sustainable Management and Technology
- College of Humanities (CDH): Digital Humanities

The EPFL programs are certified by the Swiss agency for accreditation and quality assurance (AAQ), and those in engineering have also been certified by the French Commission des Titres d'Ingénieur (CTI). Thanks to this double certification, EPFL degrees receive the label EUR-ACE and are recognized throughout Europe.

Technology transfer

EPFL acts as a start-up incubator to assure the technology transfer from academic results into industrial development. During the last decade, EPFL scientists have created an average of one new business every month. The EPFL Innovation Park hosts about 160 companies including 25 large international groups and over 130 start-ups, SME and service companies.

International Relations

While preserving the coherence of rigorous scientific training, EPFL offers its students exchange possibilities thanks to agreements with more than 200 carefully selected partner universities worldwide. EPFL is also a member of several academic networks, such as EuroTech, RESCIf and CESAER, which promote scientific collaborations as well as student and faculty mobility.

2.4 Name and status of institution (if different from 2.3) administering studies

Courses in management, marketing or economics can be chosen at the Ecole des HEC (University of Lausanne).

2.5 Languages of instruction/examination

In the grade sheet, the subject titles are referred to in the language of instruction and examination, followed by the English version if necessary.

3. Information concerning the level of qualification**3.1 Level of qualification**

Bachelor: First academic degree

3.2 Official duration of studies

The Bachelor comprises two stages:

- a. Foundation programme of 2 semesters = 60 ECTS credits
- b. Bachelor programme of 4 semesters = 120 ECTS credits

1 credit = 25-30 hours of workload

3.3 Admission requirements

Federally recognised, federal or cantonal "Maturité" or foreign upper secondary school leaving certificate recognised as equivalent.

4. Information concerning programme content and results gained**4.1 Mode of study**

Full-time

4.2 Programme requirements

180 credits obtained (see grade sheet)

4.3 Programme details and individual grades/marks/credits obtained

See grade sheet

4.4 Grading system and if available information concerning allocation of grades

Marking system EPFL

Excellent	6.0
Very good	5.5
Good	5.0
Satisfactory	4.5
Pass	4.0
Fail	3.5
Poor	3.0
Poor to very poor	2.5
Very poor	2.0
Not measurable	1.0

4.5 Overall classification of the qualification

No final classification attributed

Final Bachelor grade: 5.33 / 6**5. Information concerning the function of the qualification****5.1 Access to further studies**

Offers access to the corresponding field of study at Master level.

5.2 Professional status

No professional status awarded.

6. Additional information**6.1 Additional details on the individual course of the studies:****Was class representative during 2 semesters****6.2 Others sources of information****Life Sciences and Technology Section**EPFL Faculté SV
Bâtiment AI
Station 15
CH-1015 Lausanne
Tel: + 41 (0)21 6931634
Web: <http://ssv.epfl.ch/>
E-mail: ssv@epfl.ch**Registrar's office**EPFL AVP-E SAC
BP - Ecublens
Station 16
CH-1015 Lausanne
Tel: + 41 (0)21 693 43 45
Web: <https://www.epfl.ch/education/studies/>
E-mail: student.services@epfl.chENIC network: <http://www.enic.ch>
swissuniversities: <http://www.swissuniversities.ch>

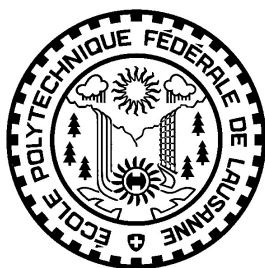
7. Endorsement of the Diploma Supplement

July 24th, 2022

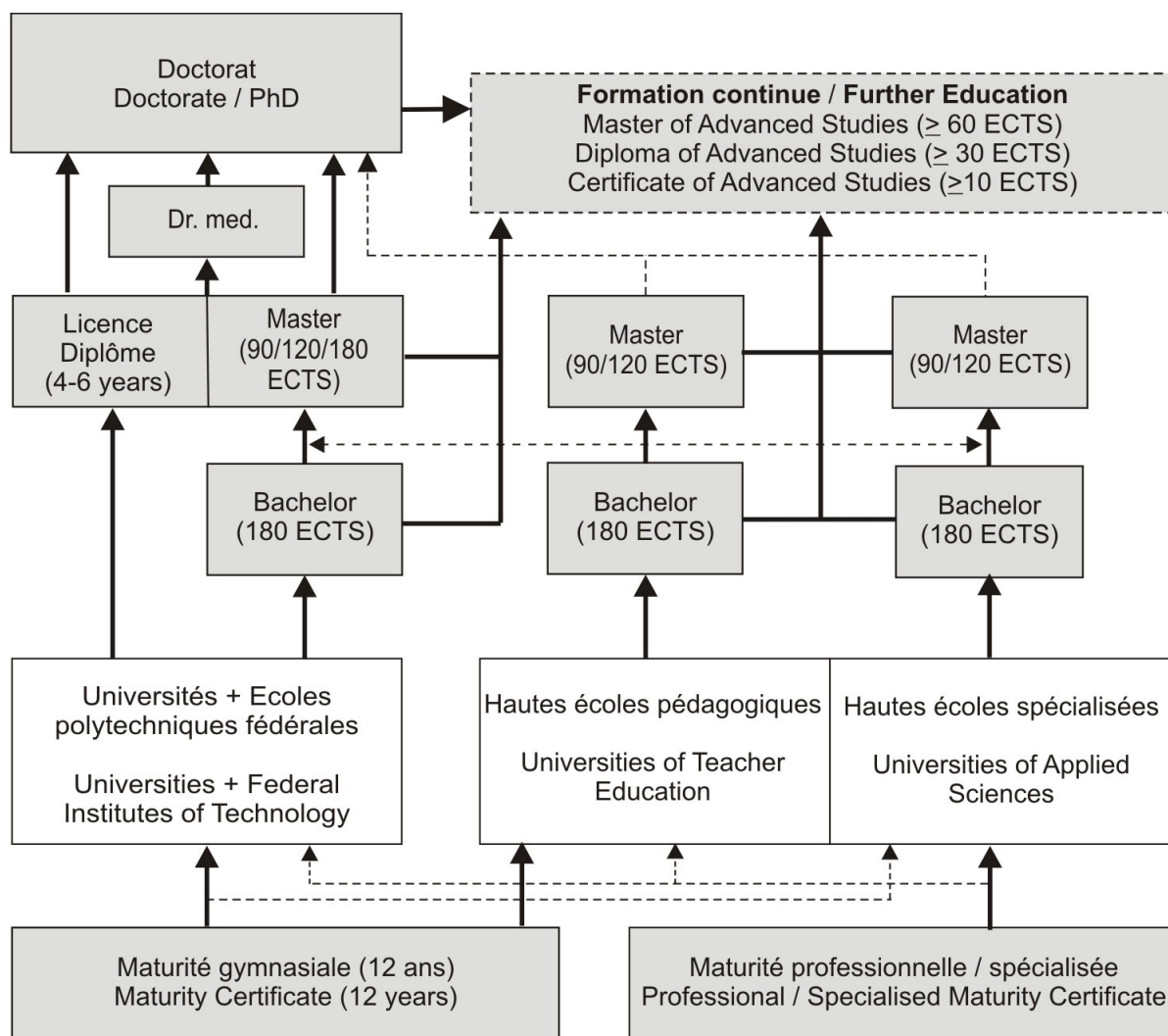
Associate Vice President for Education



Professor Pierre Dillenbourg



8. Information on the Swiss education system



→ *Parcours normal*
Normal path

-----> *Conditions supplémentaires requises*
Additional achievement required

9. Description of courses attended**BIO-204 Integrated labo in Life sciences II**

ECTS Credits : 3.

Lecturers : Grisoni-Neupert Barbara. Semester : 2020-2021 Bachelor semester 4

Students apply basic techniques in molecular biology to clone a cDNA of interest into an expression plasmid in order to produce its protein product in mammalian cells. They purify the recombinant protein and characterize it biochemically.

Les étudiants appliquent des techniques de base en biologie moléculaire pour cloner un cDNA d'intérêt dans un plasmide d'expression afin de produire la protéine correspondante dans des cellules de mammifères. Ils purifient la protéine recombinante et la caractérisent biochimiquement.

BIO-205 Cellular and molecular biology I

ECTS Credits : 3.

Lecturers : Oricchio Elisa, Suter David Michael. Semester : 2020-2021 Bachelor semester 3

The course covers the regulation of gene expression, which translates the information contained in the genome into function, by adjusting the levels and activities of mRNAs and proteins to the needs of specific cells, tissues and environments. A particular emphasis is given on experimental methods.

Le cours couvre la régulation de l'expression des gènes, qui transforment l'information contenue dans le génome en fonction, en ajustant les niveaux et les activités d'ARNm et de protéines aux demandes des cellules, tissus et environnements. Un accent particulier est mis sur les méthodes expérimentales.

BIO-207 Cellular and Molecular Biology II

ECTS Credits : 3.

Lecturers : Meylan Etienne, van der Goot Grunberg Françoise Gisou. Semester : 2020-2021 Bachelor semester 4

This course is aimed to familiarize students with the 3D organization of a eukaryotic cell, its compartmentalization, how cellular compartments communicate together and how a cell communicates with its environment. The related molecular mechanisms will be discussed.

Ce cours est destiné à familiariser les étudiants avec l'organisation 3-dimensionnelle d'une cellule eucaryote, sa compartimentalisation, comment des compartiments communiquent entre eux et comment une cellule communique avec son environnement. Les mécanismes moléculaires sous-jacents seront abordés.

BIO-212 Biological chemistry I

ECTS Credits : 3.

Lecturers : D'Angelo Giovanni, Ferreira De Sousa Correia Bruno Emanuel. Semester : 2020-2021 Bachelor semester 3

Biochemistry is a key discipline in the Life Sciences. Biological Chemistry I and II are two tightly interconnected courses that aim to understand in molecular terms the processes that make life possible.

La biochimie est une discipline clé dans les sciences de la vie. La chimie biologique I et II sont deux cours étroitement interconnectés qui visent à comprendre en termes moléculaires les processus qui rendent la vie possible.

BIO-213 Biological chemistry II

ECTS Credits : 3.

Lecturers : D'Angelo Giovanni, Ferreira De Sousa Correia Bruno Emanuel. Semester : 2020-2021 Bachelor semester 4

Biochemistry is a key discipline in the Life Sciences. Biological Chemistry I and II are two tightly interconnected courses that aim to understand in molecular terms the processes that make life possible.

La biochimie est une discipline clé dans les sciences de la vie. La chimie biologique I et II sont deux cours étroitement interconnectés qui visent à comprendre en termes moléculaires les processus qui rendent la vie possible.

BIO-311 Neuroscience

ECTS Credits : 4.

Lecturers : Mathis Mackenzie, Schneggenburger Ralf. Semester : 2021-2022 Bachelor semester 5

The course starts with fundamentals of electrical - and chemical signaling in neurons. Students then learn how neurons in the brain receive and process sensory information, and how other neurons control the behavior of an animal. Furthermore, memory, learning, and brain disorders will be introduced.

Nous commençons avec la signalisation électrique et chimique des neurones. Après, l'étudiant apprendra comment les neurones dans le cerveau reçoivent des informations sensoriels, et comment d'autres neurones contrôlent le comportement d'un animal. Finalement, la mémoire, l'apprentissage, et des ma

BIO-322 Introduction to machine learning for bioengineers

ECTS Credits : 4.

Lecturers : Brea Johanni Michael. Semester : 2021-2022 Bachelor semester 5

Students understand basic concepts and methods of machine learning. They can describe them in mathematical terms and can apply

them to data using a high-level programming language (julia/python/R).

Les étudiants comprennent les concepts et méthodes de base d'apprentissage automatique. Ils savent les décrire en termes mathématiques et ils savent les appliquer aux données en utilisant un langage de haut niveau (julia/python/R).

BIO-341 Dynamical systems in biology

ECTS Credits : 4.

Lecturers : Naef Felix. Semester : 2021-2022 Bachelor semester 5

This course introduces dynamical systems theory for modeling simple biological networks, e.g. gene circuits or population dynamics. Qualitative analysis of non-linear dynamical models will be developed together with simulations. The focus is on biological examples.

Ce cours introduit les systèmes dynamiques pour modéliser des réseaux biologiques simples. L'analyse qualitative de modèles dynamiques non-linéaires est développée de pair avec des simulations numériques. L'accent est mis sur les exemples biologiques.

BIO-372 Microbiology

ECTS Credits : 4.

Lecturers : McKinney John. Semester : 2021-2022 Bachelor semester 6

This course will provide an introduction to fundamental concepts in microbiology. Special emphasis will be given to the surprising and often counter-intuitive physical world inhabited by microorganisms.

Ce cours fournira une introduction aux concepts fondamentaux de la microbiologie. Une attention particulière sera accordée au monde physique surprenant et souvent contre-intuitif habité par des micro-organismes.

BIO-373 Genetics and genomics

ECTS Credits : 4.

Lecturers : Deplancke Bart, Fellay Jacques. Semester : 2021-2022 Bachelor semester 5

The theoretical part of this course covers classical genetics and contemporary genomics. Because the programming language R has become important for genomic research, the course also includes an introduction to R, followed by practical applications to genomic analyses, including a group project.

La partie théorique du cours couvre la génétique classique et la génomique contemporaine. Le cours comprend également une introduction au langage de programmation R, important pour la recherche en génomique, suivie d'applications pratiques aux analyses génomiques, y compris un projet de groupe.

BIO-377 Physiology by systems

ECTS Credits : 4.

Lecturers : Roy Sylvain. Semester : 2021-2022 Bachelor semester 6

The course aims at knowing and understanding the function of the cardiovascular, urinary, respiratory, gastrointestinal, basal metabolism and its regulation in the frame of a biomedical and scientific approach.

Le but est de connaître et comprendre le fonctionnement des systèmes cardiovasculaire, urinaire, respiratoire, digestif, ainsi que du métabolisme de base et sa régulation afin de développer une réflexion scientifique biomédicale.

BIO-379 Physiology lab II

ECTS Credits : 3.

Lecturers : Pexieder André. Semester : 2021-2022 Bachelor semester 6

The physiology laboratory introduces the experimental methods in the biomedical field, including experimental set-ups, transducers, signal conditioning, data acquisition and processing. The physiological end results illustrate the contents of the Physiologie par systèmes course.

Le TP de physiologie introduit les approches expérimentales du domaine biomédical, avec les montages de mesure, les capteurs, le conditionnement des signaux, l'acquisition et traitement de données. Les résultats physiologiques finaux illustrent le contenu du cours de Physiologie par système.

BIOENG-110 General Biology

Coefficient : 6.

Lecturers : Gräff Johannes, Radtke Freddy, Schoonjans Kristina, Simanis Viesturs. Semester : 2019-2020 Bachelor semester 2

The purpose of the course is to provide a general overview of the biology of cells and organisms. We will discuss this in the context of cell and organ life, focusing on the regulatory principles you will encounter in your biology studies.

Le but du cours est de fournir un aperçu général de la biologie des cellules et des organismes. Nous en discuterons dans le contexte de la vie des cellules et des organismes, en mettant l'accent sur les principes de réglementation que vous rencontrerez dans vos études de biologie.

BIOENG-312 Fluid mechanics (for SV)

ECTS Credits : 4.

Lecturers : Stergiopoulos Nikolaos. Semester : 2020-2021 Bachelor semester 4

This introductory course on fluids mechanics presents the basic concepts in fluids statics, dynamics and kinematics. All the concepts

required to take the cardiovascular track in the Bioengineering Master program are covered.

Ce cours d'introduction à la mécanique des fluides présente les connaissances de base en statique, dynamique et cinétiques des fluides nécessaires pour aborder au Master la bioingénierie du système cardiovasculaire.

BIOENG-390 Bachelor project in Life sciences

ECTS Credits : 6.

Lecturers : Hess Bellwald Kathryn. Semester : 2021-2022 Bachelor semester 6

The student will engage in a laboratory-based project in the field of life sciences engineering. Student projects will emphasize acquisition of practical skills in experimentation and data analysis. Students will acquire skills in information literacy.

L'étudiant-e s'engage dans un projet au laboratoire dans le domaine des sciences de la vie ou de l'ingénierie. Les projets mettent l'accent sur l'acquisition de compétences pratiques en laboratoire et l'analyse de données expérimentales. Les étudiants acquièrent des compétences informationnelles.

CH-112 Organic chemistry

Coefficient : 5.

Lecturers : Zakeeruddin Shaik Mohammed. Semester : 2019-2020 Bachelor semester 2

The objective of this course is to provide the knowledge and the way of understanding, at the molecular level, of the functioning of organic chemical reactions. During the course the student will think and solve new problems.

L'objectif de ce cours est de fournir les connaissances et le moyen de comprendre, au niveau moléculaire, le fonctionnement des réactions chimiques organiques. Pendant le cours, l'étudiant réfléchira et résoudra de nouveaux problèmes.

CH-160(e) Advanced general chemistry

Coefficient : 6.

Lecturers : Terrettaz Samuel. Semester : 2019-2020 Bachelor semester 1

This course aims at the acquisition of essential notions on the structure of matter, chemical equilibria and reactivity. Theoretical teaching and exercise provide the methodology to analyze and solve by reasoning and calculation novel problems of general chemistry.

Cet enseignement vise l'acquisition des notions essentielles relatives à la structure de la matière, aux équilibres et à la réactivité chimiques. Le cours et les exercices fournissent la méthodologie permettant de résoudre par le raisonnement et le calcul des problèmes inédits de chimie générale.

CS-112(i) Object oriented programming

Coefficient : 5.

Lecturers : Sam Jamila. Semester : 2019-2020 Bachelor semester 2

The fundamentals of procedural programming will be complemented with more advanced features. Emphasis will be put on object-oriented concepts, for a better reliability and reusability of programs. This course will also present some algorithmic fundamental concept.

L'objectif de ce cours est de développer des compétences de base en programmation. L'accent sera mis sur l'approche orientée-objet laquelle renforce la réutilisabilité, la maintenabilité et la fiabilité des programmes. Des fondamentaux d'algorithmique seront également abordés.

CS-116 Project in Informatics

ECTS Credits : 4.

Lecturers : Rougemont Jacques. Semester : 2020-2021 Bachelor semester 3

We introduce and put into practice principles of collaborative software development in C++ using industry-standard tools. Specific cases from life sciences are proposed and students define and implement a solution, and make it evolve throughout the semester.

On présente et on met en pratique les principes de développement en équipe d'un projet logiciel complexe en C++ à l'aide d'outils utilisés dans l'industrie. Des exemples issus des sciences de la vie sont proposés et les étudiants développent une solution et font évoluer le projet durant le semestre

CS-119(g) Information, Computation, Communication

Coefficient : 6.

Lecturers : Jobstmann Barbara, Sam Jamila. Semester : 2019-2020 Bachelor semester 1

The objective of this course is to introduce students to algorithmic thinking, to familiarize them with the fundamentals of computer and communications sciences and to develop their first programming skills (in C++).

L'objectif de ce cours est d'initier les étudiants à la pensée algorithmique, de les familiariser avec les fondamentaux de l'informatique et des communications et de développer une première compétence en programmation (langage C++).

EE-295 Electrical systems and electronics I

ECTS Credits : 4.

Lecturers : Guiducci Carlotta, Koukab Adil, Meinen Cédric. Semester : 2021-2022 Bachelor semester 5

The course has the aim to provide the fundamental knowledge and experiences to understand basic electrical systems.
Le but de ce cours est d'apporter les connaissances et les expériences fondamentales pour comprendre les systèmes électriques et électroniques de base.

EE-296 Electrical systems and electronics II

ECTS Credits : 4.

Lecturers : Koukab Adil, Meinen Cédric. Semester : 2021-2022 Bachelor semester 6

The basic concepts enabling to understand and analysis the electronic systems dedicated to signals acquisition and processing (e.g. biosensors, physiological signals, ECG) are addressed. This includes the acquisition, amplification and filtering.

Les concepts de base permettant de comprendre et d'analyser les systèmes électroniques dédiés à l'acquisition et au traitement des signaux (signaux physiologique, bio-capteurs) seront abordés en théorie et en pratique. Cela englobe l'amplification, le filtrage et l'acquisition des ces signaux.

HUM-121(b) Global issues: climate B

Coefficient : 2.

Lecturers : Randin Christophe François William, Thalmann Philippe. Semester : 2019-2020 Bachelor semester 2

This course presents the global issue of climate change: climate system and prediction; impacts on ecosystems and biodiversity; historic frame and public debate; climate targets and policies. The interdisciplinary approach introduces students to working in teams and to scientific methodologies.

Le cours présente les enjeux mondiaux liés au climat: système climatique et prévisions ; impacts sur écosystèmes et biodiversité ; cadrage historique et débat public ; objectifs et politiques climatiques. L'approche interdisciplinaire initie au travail de groupe et à la méthodologie scientifique.

HUM-215 Mediterranean myths and founding texts A

ECTS Credits : 2.

Lecturers : Bouvier David, Capponi Matteo. Semester : 2020-2021 Bachelor semester 3

The course explores the myths and the main texts of ancient cultures (Biblical World, Greece, Rome, Egypt). Exploring their meanings, their receptions, it helps us to better understand the heterogeneous nature of our own culture.

Le cours explore les mythes et les textes fondateurs des cultures antiques (Monde biblique, Grèce, Rome, Egypte). Explorant leurs significations, leurs réceptions, il s'agit de dérouler une histoire de l'imaginaire qui nous aide à mieux comprendre le caractère hétéroclite de notre propre culture.

HUM-237 Mediterranean myths and founding texts B

ECTS Credits : 2.

Lecturers : Bouvier David, Nihan Christophe. Semester : 2020-2021 Bachelor semester 4

The course explores the myths and main texts of ancient cultures (Biblical World, Greece, Rome, Egypt). Exploring their meanings, their receptions helps us to better understand the heterogeneous nature of our own culture.

Le cours explore les mythes et les textes fondateurs des cultures antiques (Monde biblique, Grèce, Rome, Egypte). Explorant leurs significations, leurs réceptions, il s'agit de dérouler une histoire de l'imaginaire qui nous aide à mieux comprendre le caractère hétéroclite de notre propre culture.

HUM-327 Industrial design D

ECTS Credits : 2.

Lecturers : Král Tomás. Semester : 2021-2022 Bachelor semester 6

Practical teaching that allows acquiring the bases of industrial design by confronting the students with the creation of an object.

Le cours propose d'expérimenter les spécificités du design industriel en confrontant les étudiant.e.s à la création d'un objet. Il s'agit d'un enseignement pratique qui repose sur le développement d'un projet de semestre.

HUM-378 Creativity and participatory design

ECTS Credits : 2.

Lecturers : Aeberli Marius, Missonier Stéphanie. Semester : 2021-2022 Bachelor semester 5

The course brings together EPFL students through a participatory design approach and reflective practice on interdisciplinary collaboration. Students develop, in teams, a prototype of an object, product, service or experience.

Le cours réunit des étudiants de l'EPFL et de l'UNIL au travers d'une approche de design participatif et d'une pratique réflexive portant sur la collaboration interdisciplinaire. Les étudiants développent, en équipe, un prototype d'objet, de produit, de service ou d'expérience.

MATH-101(f) Analysis I

Coefficient : 6.

Lecturers : Favi Giordano. Semester : 2019-2020 Bachelor semester 1

We study the fundamental concepts of analysis, calculus and the integral of real-valued functions of a real variable.

Étudier les concepts fondamentaux d'analyse et le calcul différentiel et intégral des fonctions réelles d'une variable.

MATH-106(f) Analysis II

Coefficient : 6.

Lecturers : Radici Emanuela. Semester : 2019-2020 Bachelor semester 2

The course studies fundamental concepts of analysis and the calculus of functions of several variables.

Étudier les concepts fondamentaux d'analyse et le calcul différentiel et intégral des fonctions réelles de plusieurs variables

MATH-111(f) Linear Algebra

Coefficient : 6.

Lecturers : Hess Bellwald Kathryn. Semester : 2019-2020 Bachelor semester 1

The purpose of the course is to introduce the basic notions of linear algebra and its applications.

L'objectif du cours est d'introduire les notions de base de l'algèbre linéaire et ses applications.

MATH-203(a) Analysis III

ECTS Credits : 4.

Lecturers : Dalang Robert. Semester : 2020-2021 Bachelor semester 3

This course is an introduction to the fundamental concepts of vector analysis and complex analysis with a view of their use in other courses and in solving multidisciplinary problems of scientific engineering.

Ce cours est une introduction aux concepts fondamentaux de l'analyse vectorielle et l'analyse complexe en vue de leur utilisation dans d'autres cours et pour résoudre des problèmes pluridisciplinaires d'ingénierie scientifique.

MATH-207(a) Analysis IV

ECTS Credits : 4.

Lecturers : Picasso Marco. Semester : 2020-2021 Bachelor semester 4

This course is an introduction to complex analysis and Laplace transform.

Ce cours est une introduction à l'analyse complexe et la transformée de Laplace.

MATH-231 Probability and statistics I

ECTS Credits : 4.

Lecturers : Goldstein Darlene. Semester : 2020-2021 Bachelor semester 3

Introduction to notions of probability and basic statistics.

Introduction aux notions de probabilités et les bases de la statistique.

MATH-236 Probability and statistics II

ECTS Credits : 4.

Lecturers : Goldstein Darlene. Semester : 2020-2021 Bachelor semester 4

Linear statistical methods, analysis of experiments, logistic regression.

Méthodes linéaires, analyse des expériences, régression logistique.

MATH-251(c) Numerical analysis

ECTS Credits : 4.

Lecturers : Deparis Simone. Semester : 2020-2021 Bachelor semester 4

This course introduces and analyzes numerical methods for solving mathematical problems such as linear and nonlinear systems, approximation of functions, integration and differentiation, as well as differential equations.

Le cours présente des méthodes numériques pour la résolution de problèmes mathématiques comme des systèmes d'équations linéaires ou non linéaires, approximation de fonctions, intégration et dérivation, équations différentielles.

MICRO-310(b) Signals and systems I (for SV)

ECTS Credits : 4.

Lecturers : Unser Michaël. Semester : 2021-2022 Bachelor semester 5

Introduction of the basic concepts and mathematical tools for the analysis and characterization of signals, the design of processing algorithms, and the linear modeling of systems for students in the life sciences. Application of these techniques to signal processing and communications.

Présentation des concepts et des outils de base pour l'analyse et la caractérisation des signaux, la conception de systèmes de traitement et la modélisation linéaire de systèmes pour les étudiants en sciences de la vie. Application de ces techniques au traitement et à la transmission de signaux.

MICRO-311(b) Signals and systems II (for SV)

ECTS Credits : 4.

Lecturers : Unser Michaël. Semester : 2021-2022 Bachelor semester 6

This course is an introduction to the theory of discrete linear time invariant systems. Their properties and fundamental characteristics are discussed as well as the fundamental tools that are used to study and design them (Fourier transform, Z transform).

Ce cours aborde la théorie des systèmes linéaires discrets invariants par décalage (LID). Leurs propriétés et caractéristiques fondamentales y sont discutées, ainsi que les outils fondamentaux permettant de les étudier (transformée de Fourier et transformée en Z).

MICRO-330 Sensors

ECTS Credits : 5.

Lecturers : Boero Giovanni, Renaud Philippe. Semester : 2021-2022 Bachelor semester 6

Understand the physical principles of sensors. Overview of the transduction principles and associated electronics. Examples of sensors.

Comprendre les principes physiques utilisés dans les capteurs. Vue générale des différents principes de transduction et de l'électronique associée. Montrer des exemples d'application.

PHYS-101(h) General physics : mechanics

Coefficient : 6.

Lecturers : Gruetter Rolf. Semester : 2019-2020 Bachelor semester 1

The course establishes the basic notions that will allow the student to have a better understanding of the physical phenomena in mechanics. The students acquires the capability to quantitatively analyse the consequences of these effects with appropriate theoretical tools.

Le but du cours de physique générale est de donner à l'étudiant les notions de base nécessaires à la compréhension des phénomènes physiques. L'objectif est atteint lorsque l'étudiant est capable de prévoir quantitativement les conséquences de ces phénomènes avec des outils théoriques appropriés.

PHYS-106(h) General physics : thermodynamics

Coefficient : 6.

Lecturers : Furno Ivo. Semester : 2019-2020 Bachelor semester 2

Give the student the basic notions that will allow him or her to have a better understanding of physical phenomena, such as the mechanic of point masses. Acquire the capacity to analyse quantitatively the consequences of these effects with appropriate theoretical tools.

Le but du cours de Physique générale est de donner à l'étudiant les notions de base nécessaires à la compréhension des phénomènes physiques. L'objectif est atteint lorsque l'étudiant est capable de prévoir quantitativement les conséquences de ces phénomènes avec des outils théoriques appropriés.

PHYS-201(a) General physics : electromagnetism

ECTS Credits : 6.

Lecturers : Fasoli Ambrogio, Loizu Cisse Joaquin. Semester : 2020-2021 Bachelor semester 3

The topics covered by the course are concepts of electromagnetism and electromagnetic waves.

Le cours traite des concepts de l'électromagnétisme et des ondes électromagnétiques.

PHYS-207(a) General physics : quanta

ECTS Credits : 4.

Lecturers : Savona Vincenzo. Semester : 2020-2021 Bachelor semester 4

The lecture introduces the basic concepts of quantum physics by means of experiences and relevant theories (photoelectric effect, etc.)

Le cours introduit les concepts de base de la physique quantique à l'aide d'expériences et des théories pertinentes (effet photoélectrique, etc.).